

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT1 — UNCERTAINTY VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Uncertainty Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Analyze and construct visualizations that communicate uncertainty using frequency framing, confidence intervals, trend uncertainty bands, HOPs, bootstrapping, and appropriate axis scales.(BL4)</i>		
Student Name:	Jothika M	Reg. No.:	192324235
Section / Batch:	SLOT C	Date:	27/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Show all supporting calculations (e.g., confidence interval bounds, residual standard deviation, bootstrap resample means) where relevant.
- Every diagram must include a title, correctly labeled axes, and a legend where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – UNCERTAINTY VISUALIZATION LAB

Problem 31 (5 Marks) Data: A report states: “There is a 82% chance of a student passing an exam on the first attempt.”

Task: Reframe this probability as a natural frequency statement, and describe how you would build a 10×10 icon array to represent it.

Problem 32 (5 Marks) Data: Method A vs. Method B success rate: $57\% \pm 4\%$ vs. $53\% \pm 4\%$ (95% confidence interval). Task: Calculate both confidence intervals, determine whether the two overlap, and state whether the higher figure represents a statistically meaningful lead.

Problem 33 (5 Marks) Data: monthly support tickets closed over 6 periods: 75, 85, 83, 92, 101, 102. Task: Fit a linear trend line, compute the residuals and their standard deviation, and state the 95% uncertainty band width around the trend.

Problem 34 (5 Marks) Data: A call center's daily call volume has a true mean of 620 calls and a standard deviation of 35 calls. Task: List 5 plausible individual draws that could each form one frame of a Hypothetical Outcome Plot, and explain what a viewer learns from watching all 5 frames in sequence versus a single error bar.

Problem 35 (5 Marks) Data: A small sample of 5 delivery distances (km): 33, 42, 51, 53, 54. Task: Perform one bootstrap resample (draw 5 values with replacement) and calculate its mean, then explain how repeating this process thousands of times and taking the middle 95% of resulting means gives a confidence inter

Rubrics for Calculation**ASSESSMENT RUBRIC**
AT1 — Uncertainty Visualization Lab

CO5 · BT4: Analyze ·

This rubric is used to assess student responses to the CO5-AT1 Uncertainty Visualization Lab problems, in which students construct and interpret uncertainty visualizations (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, and axis scale choice) from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant uncertainty-visualization technique (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, or axis scale choice) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, including all required calculations.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and calculation techniques to produce a complete, accurate uncertainty visualization correctly matched to the data.	Applies techniques to produce a correct visualization, with only minor errors.	Limited application; the visualization partially represents the uncertainty but has notable errors.	Fails to apply appropriate techniques; the visualization does not correctly represent the uncertainty.
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		with coherent overall flow.			
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 31 — Natural Frequencies and Icon Array (5 Marks)

Data

The report states:

“There is a 82% chance of a student passing an exam on the first attempt.”

Answer

A probability of **82%** can be converted into a natural frequency by considering **100 students**.

$$82\% = 100 \times 0.82$$

Therefore:

- **82 out of 100 students** are expected to pass the exam on their first attempt.
- **18 out of 100 students** are expected not to pass on their first attempt.

10 × 10 Icon Array

A 10 × 10 icon array contains:

$$10 \times 10 = 100$$

icons.

To represent 82%:

- Total icons = **100**
- Passing students = **82 icons**
- Not passing = **18 icons**

The first **82 icons** can be shown as the "Pass" category and the remaining **18 icons** as the "Not Pass" category.

Interpretation

The icon array makes the probability easier to understand because instead of saying **0.82 probability**, we can visually see that approximately **82 out of every 100 students** would pass on their first attempt.

R Code

R

Natural frequency for 82% probability

```
probability <- 0.82  
total_students <- 100
```

```
pass <- probability * total_students  
fail <- total_students - pass
```

```
cat("Students expected to pass:", pass, "\n")  
cat("Students expected not to pass:", fail, "\n")
```

10 x 10 icon array

```
result <- c(rep("Pass", 82), rep("Not Pass", 18))
```

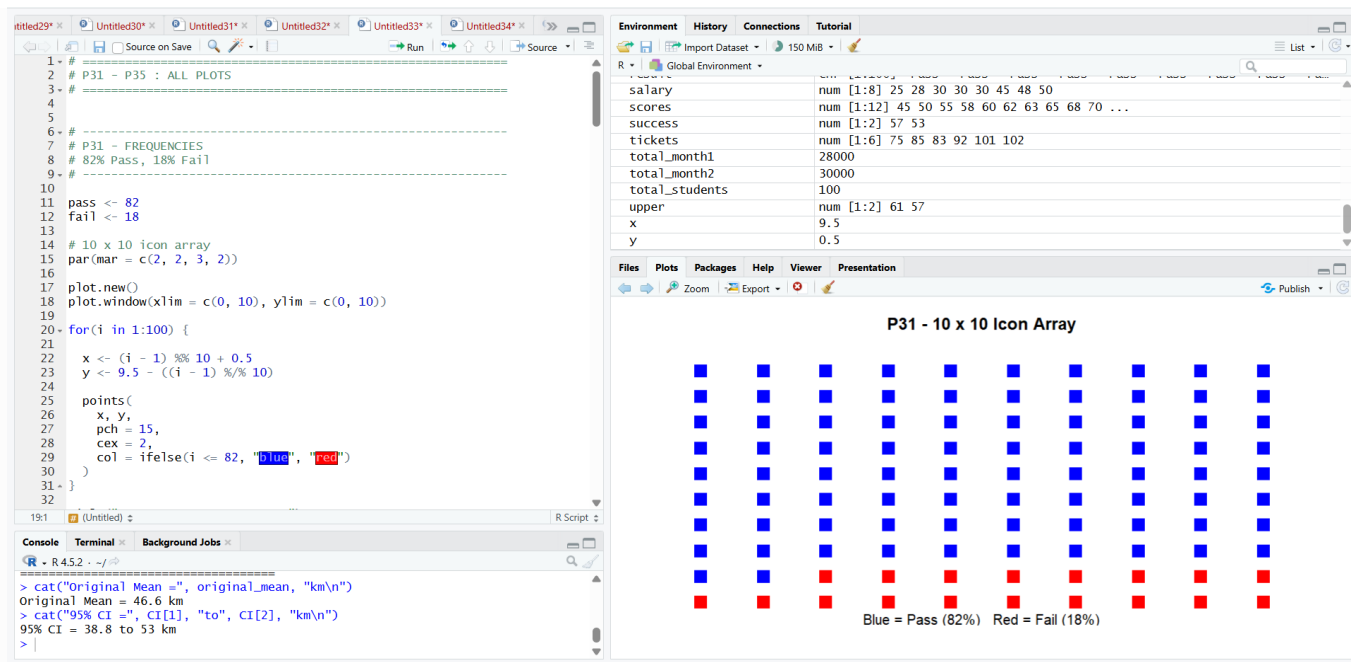
```
matrix_icons <- matrix(result, nrow = 10, ncol = 10, byrow = TRUE)
```

```
print(matrix_icons)
```

Plot icon array

```
plot(  
  rep(1:10, each = 10),  
  rep(10:1, 10),  
  pch = 15,  
  cex = 2,  
  col = ifelse(matrix_icons == "Pass", "blue", "gray"),  
  xlab = "",  
  ylab = "",  
  main = "82% Probability - 10 x 10 Icon Array",  
  xaxt = "n",  
  yaxt = "n"  
)
```

```
legend(  
  "bottom",  
  legend = c("Pass", "Not Pass"),  
  col = c("blue", "gray"),  
  pch = 15,  
  horiz = TRUE  
)
```



Result

82 icons represent students who pass and 18 icons represent students who do not pass.

Problem 32 — Confidence Intervals (5 Marks)

Data

Method	Success Rate	± Error
Method A	57%	4%
Method B	53%	4%

The values are given with a **95% confidence interval**.

Step 1: Calculate Method A Confidence Interval

Formula:

$$CI = \text{Estimate} \pm \text{Margin}$$

For Method A:

$$57\% \pm 4\%$$

Lower limit:

$$57-4=53\%$$

Upper limit:

$$57+4=61\%$$

Therefore:

$$CIA=(53\%,61\%)$$

Step 2: Calculate Method B Confidence Interval

$$53\%\pm 4\%$$

Lower limit:

$$53-4=49\%$$

Upper limit:

$$53+4=57\%$$

Therefore:

$$CIB=(49\%,57\%)$$

Step 3: Compare the Confidence Intervals

Method	Success Rate	95% CI
Method A	57%	53% – 61%
Method B	53%	49% – 57%

The intervals overlap between:

$$53\% \text{ and } 57\%$$

Therefore, **the confidence intervals overlap.**

Step 4: Is Method A's Higher Value Statistically Meaningful?

Method A has a higher observed success rate:

$$57\%-53\%=4\%$$

However, the supplied 95% confidence intervals overlap.

Therefore, based on these intervals alone, **we cannot claim that Method A has a statistically meaningful lead over Method B.**

Strictly speaking, overlap of confidence intervals is not by itself a definitive hypothesis test; a formal test of the difference would be needed for a precise significance conclusion.

Interpretation

Although Method A has a **4 percentage-point higher observed success rate**, the uncertainty around both estimates means that the evidence is **not sufficient to establish a statistically significant advantage** from the supplied information.

R Code

R

```
method <- c("Method A", "Method B")
success <- c(57, 53)
error <- c(4, 4)

lower <- success - error
upper <- success + error

ci <- data.frame(
  Method = method,
  Success_Rate = success,
  Lower_CI = lower,
  Upper_CI = upper
)

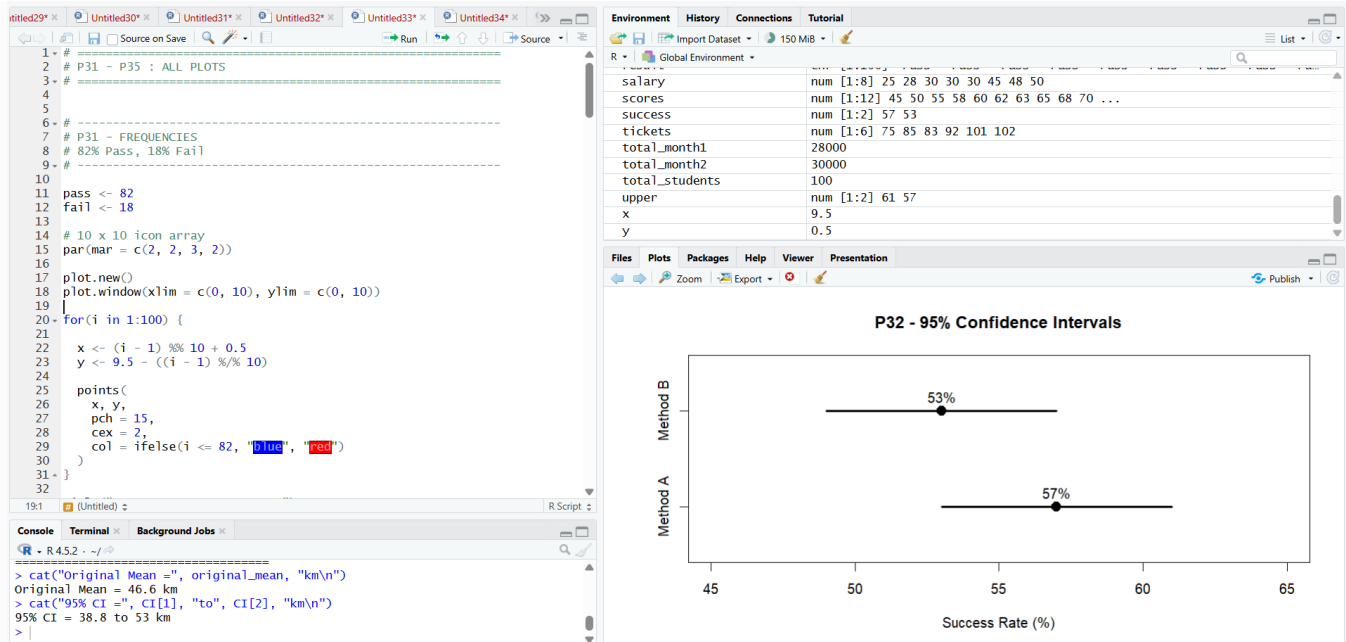
print(ci)

cat("Method A 95% CI:", lower[1], "% to", upper[1], "%\n")
cat("Method B 95% CI:", lower[2], "% to", upper[2], "%\n")

if (max(lower) <= min(upper)) {
  cat("The confidence intervals overlap.\n")
} else {
  cat("The confidence intervals do not overlap.\n")
}
```



```
cat("Observed difference:", success[1] - success[2], "percentage points\n")
```



Conclusion

Method A has a higher observed success rate, but the overlapping confidence intervals mean that a statistically meaningful lead cannot be concluded from the given information alone.

Problem 33 — Linear Trend, Residuals and Uncertainty Band (5 Marks)

Data

Monthly support tickets closed:

75, 85, 83, 92, 101, 102

Let:

$x=1,2,3,4,5,6$

and

$y=75,85,83,92,101,102$

Step 1: Fit Linear Trend Line

The linear trend equation is:

$$y=a+bx$$

The fitted equation is approximately:

$$y=70.47+5.49x$$

Therefore, the number of tickets closed increases by approximately:

$$5.49 \text{ tickets per period}$$

Step 2: Calculate Predicted Values

Period	Actual	Predicted
1	75	75.95
2	85	81.44
3	83	86.92
4	92	92.41
5	101	97.90
6	102	103.38

Step 3: Calculate Residuals

Formula:

$$Residual = Actual - Predicted$$

Period	Actual	Predicted	Residual
1	75	75.95	-0.95
2	85	81.44	3.56
3	83	86.92	-3.92
4	92	92.41	-0.41
5	101	97.90	3.10
6	102	103.38	-1.38

The residual standard deviation is approximately:

$$2.85$$

Step 4: 95% Uncertainty Band

Using the simplified 95% uncertainty calculation:

$$1.96 \times SD_{residual} 1.96 \times 2.85 \approx 5.59$$

Therefore, the trend has an approximate uncertainty of:

$$\pm 5.59$$

The total band width is approximately:

$$2(5.59) = 11.18$$

Answer

95% uncertainty band width $\approx$ 11.18 tickets, or approximately ± 5.59 tickets around the fitted trend.

This is a simplified uncertainty band based on residual standard deviation. A formal regression confidence band would vary with the x-value.

R Code

```
R
period <- 1:6
tickets <- c(75, 85, 83, 92, 101, 102)

data <- data.frame(
  Period = period,
  Tickets = tickets
)

model <- lm(Tickets ~ Period, data = data)

predicted <- predict(model)
residuals <- residuals(model)

data$Predicted <- predicted
data$Residual <- residuals

print(data)

slope <- coef(model)[2]
intercept <- coef(model)[1]

residual_sd <- sd(residuals)

uncertainty <- 1.96 * residual_sd
band_width <- 2 * uncertainty
```

```

cat("Trend equation: y =", round(intercept, 2),
    "+", round(slope, 2), "x\n")

cat("Residual standard deviation:",
    round(residual_sd, 2), "\n")

cat("95% uncertainty: +/-",
    round(uncertainty, 2), "\n")

cat("95% uncertainty band width:",
    round(band_width, 2), "\n")

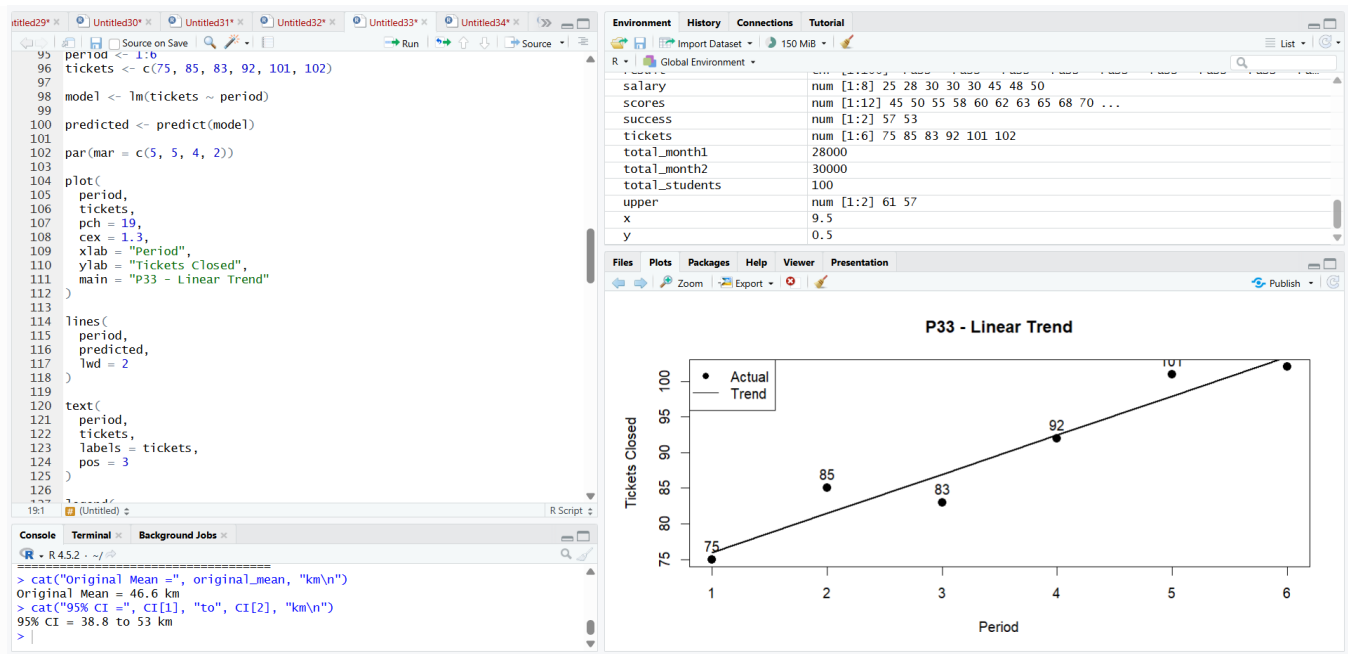
plot(
  period,
  tickets,
  pch = 19,
  type = "o",
  main = "Monthly Support Tickets with Linear Trend",
  xlab = "Period",
  ylab = "Tickets Closed"
)

abline(model, lwd = 2)

lines(
  period,
  predicted + uncertainty,
  lty = 2
)

lines(
  period,
  predicted - uncertainty,
  lty = 2
)

```



Conclusion

The support tickets show a **positive linear trend**. The fitted model indicates an increase of approximately **5.49 tickets per period**, while the residual standard deviation is approximately **2.85 tickets**.

Problem 34 — Hypothetical Outcome Plot (HOP) (5 Marks)

Data

The call center has:

- True mean = **620 calls**
- Standard deviation = **35 calls**

A plausible individual outcome can be generated around the mean.

For example, five plausible daily outcomes are:

Frame	Daily Calls
1	585
2	610
3	645

4	672
5	598

These are plausible individual draws around the mean of 620.

What Does a HOP Show?

A **Hypothetical Outcome Plot** displays individual possible outcomes one at a time.

For example:

Frame 1 → 585 calls
Frame 2 → 610 calls
Frame 3 → 645 calls
Frame 4 → 672 calls
Frame 5 → 598 calls

The viewer can see that individual daily call volumes vary even though they come from the same underlying distribution.

What Does the Viewer Learn?

Watching all 5 frames

The viewer learns that:

1. Individual outcomes are not always equal to 620.
2. Some days have fewer calls than the mean.
3. Some days have more calls than the mean.
4. Variation occurs naturally around the mean.
5. The mean represents the center of the distribution rather than every individual outcome.

Compared with a Single Error Bar

A single error bar summarizes uncertainty or variation using a static interval.

A HOP instead shows **possible individual outcomes sequentially**, making the concept of variability more intuitive.

R Code

```
R  
set.seed(123)
```

```
mean_calls <- 620
sd_calls <- 35

hop_values <- rnorm(5, mean = mean_calls, sd = sd_calls)

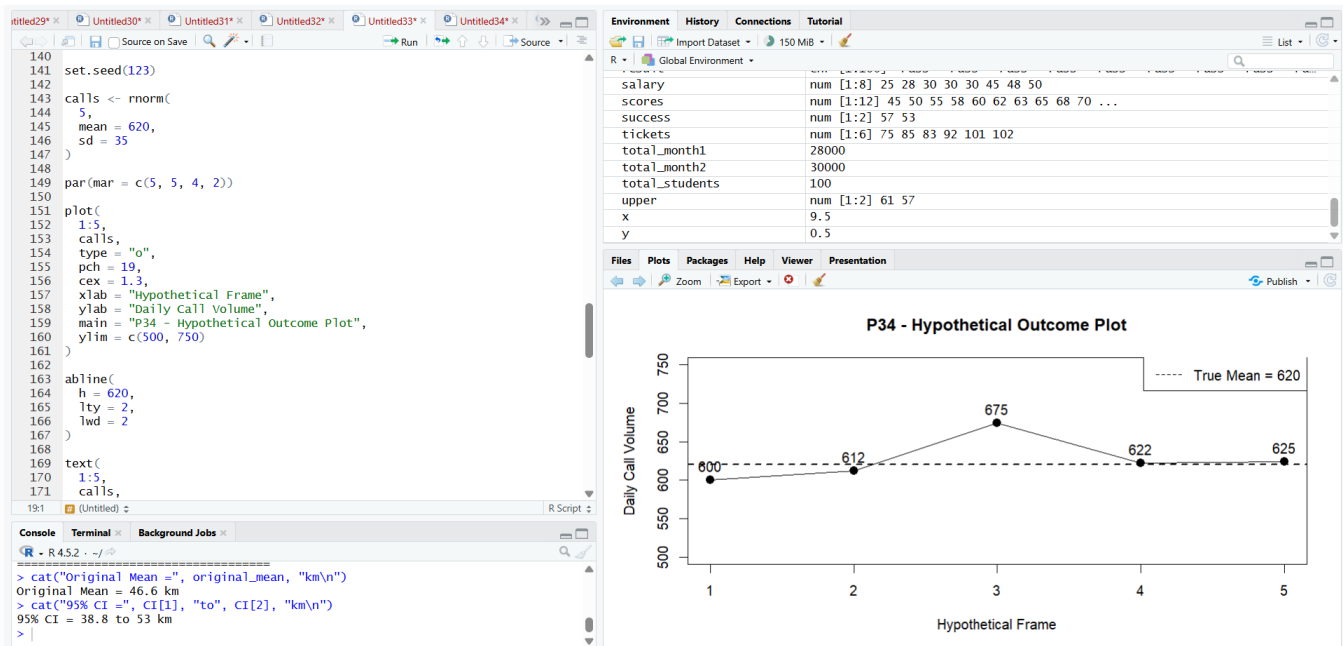
hop_values <- round(hop_values)

hop <- data.frame(
  Frame = 1:5,
  Calls = hop_values
)

print(hop)

plot(
  hop$Frame,
  hop$Calls,
  type = "o",
  pch = 19,
  main = "Hypothetical Outcome Plot",
  xlab = "HOP Frame",
  ylab = "Daily Call Volume"
)

abline(
  h = mean_calls,
  lty = 2,
  lwd = 2
)
```



Conclusion

A HOP communicates **individual outcome variability** more directly than a single error bar. Watching several frames helps the viewer understand that daily call volume can fluctuate around the true mean of 620 calls.

Problem 35 — Bootstrap Resampling (5 Marks)

Data

Five delivery distances are:

33, 42, 51, 53, 54

Step 1: Perform One Bootstrap Resample

Bootstrap sampling means drawing **5 observations with replacement** from the original sample.

One possible bootstrap sample is:

33, 54, 54, 42, 51

Notice that **54 occurs twice** and all five original values do not necessarily have to appear.

Step 2: Calculate Bootstrap Mean

$$\bar{x} = 533 + 54 + 54 + 42 + 51 = 5234 \quad \bar{x} = 46.8 \text{ km}$$

So, the mean of this particular bootstrap sample is:

46.8 km

Step 3: Repeat Thousands of Times

The bootstrap process is repeated many times.

For example:

Original sample



Randomly sample 5 values with replacement



Calculate mean



Repeat 1000s of times



Distribution of bootstrap means



Take middle 95%



95% Bootstrap Confidence Interval

Each bootstrap sample can produce a slightly different mean.

What Does the Middle 95% Mean?

After thousands of bootstrap means are obtained:

- Remove the lowest 2.5%
- Remove the highest 2.5%
- Keep the middle 95%

The lower and upper boundaries of this middle 95% form the **bootstrap confidence interval**.

This provides an estimate of the uncertainty around the population mean based on the original sample.

R Code

```
R
distances <- c(33, 42, 51, 53, 54)

set.seed(123)

bootstrap_sample <- sample(
  distances,
  size = 5,
  replace = TRUE
)

bootstrap_mean <- mean(bootstrap_sample)

cat("Original data:", distances, "\n")
cat("Bootstrap sample:", bootstrap_sample, "\n")
cat("Bootstrap mean:", bootstrap_mean, "km\n")
```

Thousands of Bootstrap Resamples

```
R
distances <- c(33, 42, 51, 53, 54)

set.seed(123)

B <- 10000

bootstrap_means <- numeric(B)

for (i in 1:B) {
  sample_data <- sample(
    distances,
    size = 5,
    replace = TRUE
  )

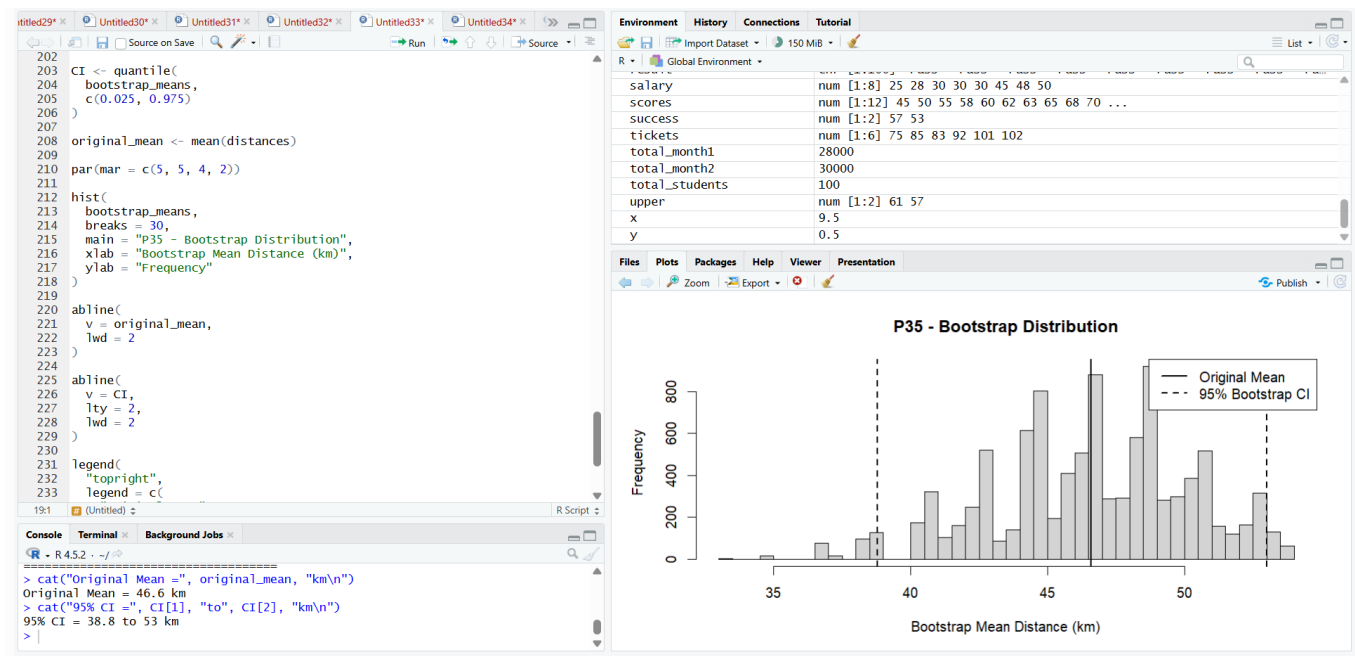
  bootstrap_means[i] <- mean(sample_data)
}

confidence_interval <- quantile(
  bootstrap_means,
  c(0.025, 0.975)
)
```

```
print(confidence_interval)
```

```
hist(  
  bootstrap_means,  
  breaks = 30,  
  main = "Bootstrap Distribution of Mean Delivery Distance",  
  xlab = "Bootstrap Mean (km)"  
)
```

```
abline(  
  v = confidence_interval,  
  lty = 2,  
  lwd = 2  
)
```



Conclusion

One possible bootstrap sample is:

33,54,54,42,51

with mean:

46.8 km

Repeating the resampling thousands of times creates a **bootstrap distribution of sample means**. The middle **95%** of those means provides an estimated **95% bootstrap confidence interval** for the population mean.